# Multi-View 3D Wheat Ear Instance Segmentation for RGB-Based Phenotyping

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**Abstract.** This work investigates individual wheat ear phenotyping from handheld mobile phone video. We combine wheat-specific 2D segmentation, 3D Gaussian Splatting (GS) reconstruction, and an SA3D-GS-based multi-view association procedure that builds persistent 3D ear instances from repeated 2D observations. We introduce an annotated dataset with frame-level masks, persistent ear identities, manual morphometric measurements, and CT-derived volumes. We evaluate 2D segmentation, 3D instance recovery, and estimation of ear length, width, and volume against physical and CT-derived references.

**Keywords:** Plant phenotyping · Wheat ears · SAM · 3DGS · SA3D-GS · Multi-view segmentation

## 1 Introduction

Accurate organ-level morphology is important for high-throughput plant phenotyping and crop breeding. Manual measurements are laborious and often destructive, while measurements from individual 2D images are inherently view-dependent. Multi-view RGB imaging offers a low-cost alternative, but reliable 3D phenotyping requires reconstruction and instance-level separation of thin, visually similar, and frequently occluded wheat ears. Recent methods transfer 2D masks or visual features into radiance-field and 3DGS representations [1,2,9,10]. For wheat phenotyping, Wheat3DGS [11] demonstrates multi-instance wheat ear segmentation and trait extraction by associating multi-view 2D masks with Gaussian primitives.

In this work, we investigate an alternative strategy based on SA3D-GS [2]. Our iterative multi-view procedure leverages GS reconstruction to match 2D-segmented wheat ears across the video sequence, reusing existing instances across views and initializing new segments only for previously unrepresented ears. SA3D-GS then back-projects the matched 2D segmentation masks into 3D to obtain instance-level wheat ear segmentation, while the iterative instance matching and initialization process automatically determines the number of wheat ears in the scene and helps resolve occlusions that can cause 2D segmentation to leak across multiple ear instances.

Our contributions are threefold: (1) a dataset with 3D morphometric measurements; (2) an extension of SA3D-GS to multi-instance segmentation of wheat ears; and (3) a comparison with Wheat3DGS, demonstrating more accurate wheat ear counts, higher recall, and more accurate estimates of morphometric traits, resulting in improved biomass estimation.

## 2 Methodology

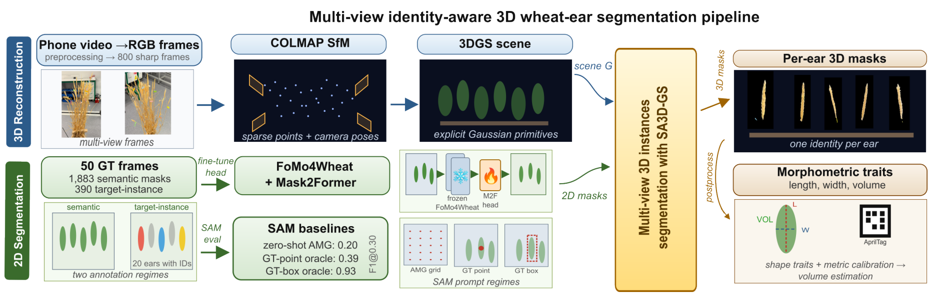


**Fig. 1:** Multi-view identity-aware 3D wheat ear segmentation pipeline: Starting from initial RGB frames (video) to individual 3D wheat ear segments and morphometric traits.

Our pipeline operates in three main stages. First, we perform 2D wheat ear segmentation in each frame using the SAM model with various prompts and a fine-tuned FoMo4Wheat model. Second, a 3D reconstruction is generated using a 3DGS model initialized from SfM. Finally, we perform 3D instance segmentation by iteratively applying SA3D-GS. This process matches back-projected segment IDs from Gaussian Splats with 2D segmentation masks across frames, enabling the identification and tracking of all individual wheat ears throughout the scene, unlike the original SA3D-GS.

**Image preprocessing and 2D segmentation.** For 2D segmentation, we fine-tuned FoMo4Wheat as a crop-domain visual foundation model [5]. The backbone was initialized from the pretrained FoMo4Wheat-Large checkpoint and kept frozen, while the Mask2Former head was fine-tuned on our training data [3,4]. We additionally evaluated the automatic and ground-truth (GT)-prompted variants of SAM [7] as diagnostic baselines. The GT-prompted variant provides the model with explicitly specified wheat ear locations, allowing us to assess its segmentation capability under ideal prompting conditions and providing an upper-bound reference for the 2D segmentation performance.

**3D reconstruction.** Camera poses and sparse geometry are estimated with COLMAP [8] and used to initialize 3D Gaussian Splatting [6]. Models are optimized for 30,000 iterations, and the reconstruction configuration is selected based on held-out views using PSNR, SSIM, and LPIPS before training the final model on all available views.

**Multi-view 3D instance segmentation.** Let $\mathcal{G} = \{g_i\}_{i=1}^{N}$ denote the reconstructed Gaussian scene and let $\mathcal{M}_t = \{M_{t,k}\}$ be the set of 2D wheat ear masks observed in the $t$-th selected camera view $v_t$, and $k$ indexes the 2D masks within a view. Given a mask $M_{t,k}$, SA3D-GS [2] produces a subset of Gaussians $\mathcal{O}_{\text{new}} = \Phi(\mathcal{G}, v_t, M_{t,k})$ corresponding to the segmented $j$-th wheat ear. Applying this operation independently for every view, however, would repeatedly segment the same physical ear. We therefore process views progressively. Each existing instance $O_j$ is rendered (denoted by $\Pi_t$) into a new view $\widehat{M}_{j,t} = \Pi_t(O_j)$ and matched against the current 2D masks using $q_{j,k} = \mathrm{IoU}(\widehat{M}_{j,t}, M_{t,k})$ where $q_{j,k}$ is the 3D-2D matching score. Candidate pairs with $q_{j,k} > 0.3$ are sorted by decreasing overlap and assigned greedily under a one-to-one constraint. Matched observations retain the existing 3D identity, whereas unmatched masks initialize new SA3D-GS segments. The resulting Gaussian masks are cleaned using 3D connected-component filtering and duplicate merging. The complete procedure is given as pseudocode in the Supplementary Material (Suppl.Mat. Sec. 1).

**Morphometric extraction.** Morphometric traits are extracted from the Gaussian centers of each accepted 3D instance, for which PCA defines the principal axes, and length and width are estimated from the 2nd–98th percentile range. Scene coordinates are converted to millimeters using an AprilTag-derived scale factor. Volume is estimated from an alpha-shape reconstruction with alpha radius set to 1.5 times the median nearest-neighbor distance.

## 3 Results

We first describe a dataset we have created, which is unique for its 3D morphometric ground-truth measurements. Since our focus is accurate extraction of these measurements, we first evaluate 3D instance segmentation and compare it to the Wheat3DGS and then present the morphometric evaluation. 2D segmentation and 3D reconstruction discussions are part of the supplementary material.

**Dataset** A single wheat plant was recorded with an iPhone 16 Pro while the camera moved around the static sample under a non-uniform laboratory background. Of 1,057 extracted frames, blur filtering retained 800 for 3D reconstruction. The dataset contains 50 manually annotated frames with all visible wheat ears (1,883 masks), and persistent identities for the 20 colored-sticker marked ears. The annotations were split into 35/5/10 train, validation, and test frames. Selected ears were subsequently measured manually and scanned by X-ray CT for morphometric validation.

**3D instance segmentation.** We evaluate our proposed pipeline adapted to SA3D-GS and Wheat3DGS on the same reconstructed scene and common evaluation protocol (Table 1). Wheat3DGS obtains higher object-level F1 and slightly higher overlap and precision for the matched instances, with 11/20 accepted target matches compared with 10/20 for our pipeline. Our extension of SA3D-GS pipeline, however, produces 38 instances, close to the average GT count of 38.2 visible ears per annotated frame, and shows fewer missed positive regions: recall is higher both over the complete prediction set (0.803 vs 0.664) and over the matched instance subset (0.966 vs 0.926).

**Table 1:** Comparison of the selected 3D instance-segmentation configurations. The average GT count is 38.2 visible ears per evaluation frame. "All" metrics use the full rendered prediction set; matched instances metrics are averaged over the ten highest-IoU one-to-one matched target ears for each method.

| | Recovery | | Object F1 | | All predictions | | | | Matched instances | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | Pred. | Match/20 | @.30 | @.50 | IoU | Dice | P | R | IoU | Dice | P | R |
| Ours | **38** | 10 | 0.345 | 0.103 | **0.313** | **0.445** | 0.322 | **0.803** | 0.477 | 0.643 | 0.484 | **0.966** |
| Wheat3DGS | 28.4 | **11** | **0.458** | **0.125** | 0.304 | 0.423 | **0.329** | 0.664 | **0.488** | **0.649** | **0.518** | 0.926 |

**Table 2:** Morphometric evaluation on the physically measured ears recovered by both methods. Length and width errors - in mm, volume - in mm$^3$, MAPE - in %.

| | Length | | | Width | | | Volume | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Method | MAE | RMSE | $r$ | MAE | RMSE | $r$ | MAE | RMSE | $r$ | MAPE |
| Ours | **4.67** | **5.88** | 0.791 | **3.70** | **5.88** | -0.420 | **305.29** | **389.46** | 0.514 | **16.35** |
| Wheat3DGS | 10.16 | 10.63 | -0.707 | 5.74 | 8.65 | -0.315 | 459.96 | 547.43 | 0.038 | 25.48 |

**Morphometric evaluation.** Table 2 compares morphometric errors on the measured ears recovered by both methods (per-ear measurements are reported in the Suppl.Mat. Sec. 6). Our estimates show positive Pearson correlations r with the reference values for length and volume. Despite Wheat3DGS's stronger matched instance segmentation scores, our pipeline gives lower errors for all three traits: length MAE is 4.67 vs 10.16 mm, width MAE is 3.70 vs 5.74 mm, and CT-referenced volume MAPE is 16.35% vs 25.48%. This indicates that instance-level segmentation scores alone do not fully characterize suitability for phenotyping. Errors in the geometric extent of a recovered 3D object can have a substantial effect on downstream measurements.

## 4 Discussion and Future Work

Our method and dataset have limitations: the dataset requires more annotated sequences covering various wheat growth stages in both lab and outdoor environments. The current method is slow due to separate SA3D-GS runs per ear, a known issue we are addressing. Future work, with more annotated data, aims for a holistic approach where 2D and 3D segmentation mutually enhance each other, further improving recall and significantly boosting precision.

## 5 Conclusion

We developed an RGB-based pipeline for 3D wheat ear segmentation and phenotyping, adapting SA3D-GS with iterative multi-view instance association. Compared to Wheat3DGS, our method achieves more accurate recovery of the total number of wheat ears, higher recall, and lower errors in length, width, and CT-referenced volume, despite lower object-level F1, overlap, and precision for matched instances. This study, limited to a single static laboratory plant, suggests future validation across diverse plants and field conditions.

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